Data Communication & Computer Networks (CN)

Topic 5 of 7 : Transmission Medium

45 Pending Questions — staged from repo exam PDFs & trusted sources — **NEED VERIFICATION** (answers + Hindi to be verified during topic build)

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)**

Total Questions: **45** pending questions (staging)

Question Type: MCQ — 4 options (A, B, C, D) per question, in English + Hindi

Layout: Left = **question** | Right = **explanation** (green): ✔ **why the correct answer is right** + ★ **Nukta** (key exam point)

Language rule: Hindi is the second language; **technical terms stay in English** (topology, protocol, jitter…)

Verification: Every answer cross-verified **4×** (Forouzan + official answer keys of Subject Test / Unit Test Solutions). Wrong options verified internally.

Syllabus Coverage — sub-topic wise order

How to read: Left side: question + 4 options (try it yourself first). Right side (green): ✔ correct answer with why it is right, and ★ **Nukta** — the one-line exam tip to remember. Practice in 2 passes: first read question + options only, then check the green column.

System: generated by **tools/cn_build_topic.py** from **tools/data/topic5_data.py**. To add more questions later: drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions auto-classify topic-wise. See **Computer Network/README.md**.

Pending Review — auto-ingested, needs verification

Auto-extracted & classified; answers to be verified before counting.

P1. "Manchester Encoding" and "Differential Manchester Encoding" are signal encoding techniques used at which layer? "Manchester Encoding" "Differential Manchester Encoding"

(A) Physical Layer
(B) Data Link Layer
(C) Network Layer
(D) Presentation Layer

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

P2. Assertion
(A) : Fiber Optic cable is immune to Electromagnetic Interference (EMI). (A): (EMI) Reason (R): It transmits data in the form of light pulses rather than electrical signals. (R): (A) Both (A) and (R) are true, and (R) is the correct explanation of (A)
(B) Both (A) and (R) are true, but (R) is NOT the correct explanation.
(C) (A) is true, but (R) is false. /.
(D) (A) is false, but (R) is true.
[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P3. A company needs long-distance communication (5–10 km), high data rate, and immunity from electromagnetic interference (EMI). Which transmission media is most suitable? (5–10), EMI transmission media
(A) UTP
(B) Coaxial cable
(C) Single-mode fiber
(D) Multi-mode fiber
[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P4. In a copper cable network, attenuation increases when frequency increases. What change will reduce attenuation? frequency attenuation attenuation
(A) Increase cable length
(B) Use lower category UTP
(C) Use thicker copper core
(D) Increase signal frequency
[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

P5. A network uses coaxial cable but suffers from signal reflection. Which problem is most likely the cause? coaxial cable signal reflection
(A) EMI
(B) Incorrect termination resistance
(C) Crosstalk
(D) Low bandwidth
[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P6. Which of the following statements are TRUE for fiber optic cables? I. Immune to tapping II. High bandwidth III. Requires electrical grounding IV. Works using total internal reflection
(A) I, II, IV
(B) II, III
(C) I, III, IV
(D) I, II
[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P7. A UTP cable used in Gigabit Ethernet shows high delay skew. Which is the cause? Gigabit Ethernet UTP delay skew

(A) Different twist rates in cable pairs
(B) Excessive EMI
(C) Fiber splice error
(D) Wrong wireless channel

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P8. A channel has high bandwidth but suffers from high noise. Which encoding scheme will provide better error resistance? bandwidth noise - encoding error resistance

(A) NRZ
(B) Manchester
(C) AMI
(D) ASK

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

P9. Which phase of a computer virus life cycle is responsible for remaining hidden until a specific condition is met? D H

(A) Propagation Phase
(B) Triggering Phase
(C) Dormant Phase / C (E)
(D) Execution Phase / F

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ auto-ingested — answer needs verification

P10. Reduction in signal strength over distance is called: v %– B

(A) Distortion –
(B) Noise –
(C) Attenuation – L
(D) Interference – T

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

P11. Which noise is caused due to electromagnetic coupling between nearby cables? ™S v - š›

(A) Thermal noise – Z
(B) Crosstalk –]
(C) Impulse noise – ^_
(D) Quantization noise – c

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

P12. Which parameter of an analog signal represents the number of cycles per second? v ‰ ‡ 9

(A) Amplitude –
(B) Phase –
(C) Frequency – k
(D) Wavelength – pZ

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ auto-ingested — answer needs verification

P13. Which impairment cannot be completely eliminated but only minimized? Ÿ B y
(A) Attenuation – L
(B) Distortion –
(C) Noise –
(D) Delay –
[Source: BCI Subject Test-9.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

P14. Which of the following is a wireless networking technology? / H I + K
(A) Ethernet
(B) Wi-Fi
(C) Fiber Optic / H
(D) DSL
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P15. What is the maximum data rate of a transmission medium known as? / ? Q
(A) Bandwidth / P
(B) Throughput / R
(C) Latency / T
(D) Jitter
[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

P16. Which cable has maximum bandwidth? ?**
(A) Coaxial
(B) Twisted pair
(C) Optical fiber
(D) USB cable
[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P17. Which wireless standard provides the highest data rate
(A) 802.11b
(B) 802.11g
(C) 802.11n
(D) 802.11ac What is the purpose of DHCP Discover message?DHCP Discover (a) Request IP lease (b) Identify gateway (c) Detect MAC address (d) Announce server
[Source: CN Topic Test 6.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ auto-ingested — answer needs verification

P18. What kind of transmission medium is most appropriate to carry data in a computer network that is exposed to electrical interferences
(A) Unshielded twisted pair
(B) Optical fiber
(C) Coaxial cable
(D) Microwave
[Source: CN Topic Test-01.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P19. Which of the following cannot be used as a medium for 802.3 ethernet? 802.3

(A) A thin coaxial cable
(B) A twisted pair cable
(C) A microwave link
(D) A fiber optical cable

[Source: CN Topic Test-01.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P20. What technique is used to allow multiple signals to share the same communication channel in a network

(A) Multiplexing
(B) Phase modulation
(C) Analog modulation
(D) Digital modulation

[Source: CN Topic Test-02.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

P21. What is the meaning of bandwidth in a network

(A) Transmission capacity of a communication channel
(B) Connected components in a network
(C) Class of IP used in Network
(D) Interconnected by communication channels

[Source: CN Topic Test-02.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

P22. The period of a signal is 100ms. What is its frequency in kilohertz? 3. 100

(A) 10-1 kHz
(B) 10 -2 kHz
(C) 10-3 kHz
(D) 10-4 kHz

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ auto-ingested — answer needs verification

P23. Which of the following is NOT a type of transmission media

(A) Coaxial cable
(B) Fiber optics
(C) NIC
(D) Twisted pair

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P24. In digital communication, what does "attenuation" refer to? " "

(A) Signal delay
(B) Signal loss
(C) Signal interference
(D) Signal amplification

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

P25. Which of the following best explains why fiber optic cables have a lower bit error rate than copper cables

(A) Higher signal power
(B) Immunity to electromagnetic interference
(C) Simpler modulation techniques
(D) Larger physical size

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P26. Given a bit rate of b bits/sec, the time required to send 16 bits is: b / , 16 required :

(A) 16*b sec
(B) 16/b sec
(C) 16b sec
(D) b16 sec

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

P27. Which of the following transmission media works on the principle of total internal reflection

(A) Optical fiber cable
(B) Shielded twisted pair cable
(C) Unshielded twisted pair cable
(D) coaxial cable

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P28. An analog signal carries 4 bits in each signal unit. if 1000 signal units are sent per second, then baud rate and bit rate of the signal are 4 1000

(A) 1000 bauds/sec, 4000 bps
(B) 1000 bauds/sec, 500 bps
(C) 4000 bauds/sec, 1000 bps
(D) 2000 bauds/sec, 1000 bps

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

P29. Crosstalk is typically minimized in twisted pair cables by:

(A) Increasing the power level
(B) Increasing the twist rate
(C) Adding shielding to the core
(D) Using longer cables

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P30. What is the typical maximum length of an unshielded twisted pair (UTP) cable in Ethernet before signal degradation occurs? ( )

(A) 50 meters
(B) 100 meters
(C) 150 meters
(D) 200 meters

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P31. Which connector is commonly used for coaxial cable in physical networks

(A) RJ-45
(B) USB-C
(C) BNC
(D) HDMI

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Guided Media]

⚠ auto-ingested — answer needs verification

P32. Which type of wireless transmission requires line-of-sight

(A) Wi-Fi
(B) Infrared
(C) Radio
(D) Bluetooth

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ auto-ingested — answer needs verification

P33. The effective bandwidth of a communication channel is primarily limited by:

(A) Protocol overhead
(B) Noise and attenuation
(C) File size
(D) MAC address filtering

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

P34. Which of the following statements is not true with respect to microwaves

(A) Electromagnetic waves with frequencies from 300 GHz to 400 Thz.
(B) Propagation is line-of-sight.
(C) Very high-frequency waves cannot penetrate walls.
(D) Use of certain portions of the band requires permission from authorities. 28 Optical fiber uses reflection to guide light through a channel, in which angle of incidence is_____the critical angle. _____ (a) A equal to (b) less than (c) greater than (d) less than or equal to

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ auto-ingested — answer needs verification

P35. Which of the following transmission media is used in Bluetooth Technology: :

(A) Radio links
(B) Microwave links
(C) CVSAT Communication
(D) Fiber-optic

[Source: CN Topic Test-04.pdf | auto-classified: Topic 5 / Unguided Media]

⚠ auto-ingested — answer needs verification

P36. The primary use of scramblers in Line Coding Schemes is to -
(A) make the data look random
(B) compress transmission data
(C) increase signal bandwidth
(D) decrease signal bandwidth
[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 5 / Signals & Performance]

⚠ auto-ingested — answer needs verification

P37. - transmission medium modern internet backbone networks primary ? Which transmission medium is primarily used in modern internet backbone networks
(A) Twisted Pair Cable
(B) Coaxial Cable
(C) Single Mode Fiber Optic Cable
(D) Microwave Links
[Source: BCI Subject Test-8.pdf | auto-classified: Topic 1 / DC Components & Protocol Basics]

⚠ auto-ingested — answer needs verification

P38. Which physical layer technique allows multiple users to share the same bandwidth simultaneously
(A) TDMA
(B) FDMA
(C) CDMA
(D) ARQ
[Source: CN Topic Test-04.pdf | auto-classified: Topic 2 / OSI Model Basics]

⚠ auto-ingested — answer needs verification

P39. signals are discrete and generated by digital modulations. Analog Digital Noise Bandwidth
(A) T
(B) n
[Source: Random Test- Rajasthan PYQ.pdf (OCR), p5 | auto-classified: Topic 5 / Signals & Performance]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

P40. CQuanizing 39. Quantizing noise occurs in Pulbre Gade Moduoechon APCM
(A) TDM
(B) FDM
(C) PPM
(D) PCM (B) TDM
[Source: Random Test- Rajasthan PYQ.pdf (OCR), p10 | auto-classified: Topic 5 / Signals & Performance]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

P41. Llhh Division Mwiple Access 83. Select the false statement about CDMA technology 益益 I.CDMA technology optimizes theuseof availablebandwidthasit transmitsover the C1:SL entire frequency range. (2:S2 ≤s:≤) II.Each user in a CDMA system uses a c4:54 different code to modulate their signal y 43 CDMA technology offers lesssecurity as compared to theGSM technology. A SIMcard is alwaysrequired for the Gmpvice workingofa CDMA device. CDMA Moc eCM Only (I) Xonly (H) Io GSM ()and (II) (Ill) and (IV)
[Source: Random Test- Rajasthan PYQ.pdf (OCR), p11 | auto-classified: Topic 5 / Unguided Media]

⚠ OCR-parse staging: verify options/answer at curation (see src page). OPTIONS INCOMPLETE in OCR — re-check source page.

P42. 45.lf the channel band is limited to 6kHz and channe signal to noise ratio is 16,what would be the capacity of channel? og(4 zgir

(A) 15.15kbps 24.74kbps Radio Noicy Nosdess (g(8)
(B) 30.12kbps
(C) 52.18kbps chann channel (dNs+) 01009 (1+16)
(D) 15.15kbps

[Source: Random Test- Rajasthan PYQ.pdf (OCR), p30 | auto-classified: Topic 5 / Signals & Performance]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

P43. 8. Which of the ‘following Ethernet encoding has the highest redundancy' Encading) 1B2B (q) 4B5B 5B6B prerent g< 8B10B abser. Fke Nw ophcad 3J59

[Source: Unit Test-4 Solution.pdf (OCR), p9 | auto-classified: Topic 5 / Signals & Performance]

⚠ OCR-parse staging: verify options/answer at curation (see src page). OPTIONS INCOMPLETE in OCR — re-check source page.

P44. 22. The amount of data transmitted in a fixed amount of time iscalled

(A) Refresh rate (b) Resolution efBandwidth
(B) Frequency

[Source: Unit Test-4 Solution.pdf (OCR), p23 | auto-classified: Topic 5 / Unguided Media]

⚠ OCR-parse staging: verify options/answer at curation (see src page).

P45. 29. The encoding technique used to transmit the signal in giga Ethernet technology over fiber optic medium is

(A) Differential Manchester encoding
(B) Non Return to zero
(C) 4B
(D) 8B

[Source: Unit Test-4 Solution.pdf (OCR), p30 | auto-classified: Topic 5 / Signals & Performance]

⚠ OCR-parse staging: verify options/answer at curation (see src page).